

CONTACT TOPOLOGY WORLD MODELS DO NOT IMPROVE MUJoCo CONTACT-CHAIN CONTROL

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ABSTRACT

We rebuilt a generated robotics idea into a concrete contact-rich manipulation study: should world models predict contact graph topology changes rather than only state trajectories? The v4 rebuild implements a MuJoCo tabletop benchmark with a pusher, two movable blocks, walls, a fixture, and a target pocket. It logs contact edges, births/deaths, graph edit distance, components, jam/safety events, task progress, seven random seeds, uncertainty intervals, paired statistics, ablations, stress sweeps, and negative cases. The result is negative. On the decisive combined-stress split, the proposed `topology_world_model` reaches 0.107 ± 0.083 task success, while the strongest non-oracle baseline, `ensemble_uncertainty_planner`, reaches 0.125 ± 0.076 . The paired success difference is -0.018 ± 0.035 . The topology model improves edge F1 over that ensemble baseline, but does not convert graph prediction into better control. We therefore mark the paper **KILL/ARCHIVE** for ICLR main.

1 DECISION

This document is the v4 ICLR-main gate for Paper 73, *Contact Topology World Models*. The previous version was killed because it had only a synthetic template. The current rebuild adds a real local MuJoCo benchmark and implemented baselines. The rejection reason has changed: the evidence now exists, but it does not support the central claim.

2 HYPOTHESIS

The tested hypothesis is that explicitly predicting contact graph topology changes, including contact births/deaths, connected components, and jam modes, improves contact-rich action selection beyond state-only dynamics, distance thresholds, pairwise contact classifiers, uncertainty ensembles, and contact-implicit control. For the paper to become submission-relevant, topology prediction must improve downstream task success, not only an intermediate graph metric.

3 BENCHMARK

The benchmark uses MuJoCo dynamics for a planar pusher, two sliding blocks, static walls, a movable fixture, and a pocket. The task is to create a useful pusher-block-block-pocket contact chain without jamming against the fixture or walls. Five evaluation splits are used: nominal push-to-pocket, contact-chain transfer, fixture topology shift, friction/mass shift, and combined stress. Combined stress includes fixture shift, friction/mass shift, contact-sensor noise, actuator limits, and distractor contacts.

The main evaluation contains 2240 rollouts: 5 splits, 7 seeds, 8 episodes per seed/split, and 8 methods. The ablation study contains 252 rollouts. The stress sweep contains 1050 rollouts. All intervals are 95 percent confidence intervals over seed-level means.

4 METHODS

The implemented methods are:

- `last_contact_persistence`: predicts the next graph from the previous graph.
- `distance_threshold_graph`: predicts contacts from pairwise distances and relative geometry.
- `state_only_dynamics_model`: predicts state deltas and induces contacts from predicted state.
- `pairwise_contact_classifier`: supervised pairwise contact predictor.
- `ensemble_uncertainty_planner`: conservative ensemble over pairwise/topology predictors.
- `contact_implicit_mpc_baseline`: geometric contact-implicit controller with contact-distance penalties.
- `topology_world_model`: proposed contact graph transition predictor and topology-aware planner.
- `oracle_topology_planner`: upper bound with true next topology for prediction metrics.

5 MAIN RESULTS

Table 1 shows the combined-stress result. The topology model has lower safety violations than several baselines and higher edge F1 than the uncertainty ensemble, but it does not win on task success.

Method	Success	Edge F1	Graph edit	Safety	Progress
<code>oracle_topology_planner</code>	0.250 ± 0.076	1.000	0.000	0.000	0.707
<code>ensemble_uncertainty_planner</code>	0.125 ± 0.076	0.364	0.276	0.237	0.649
<code>contact_implicit_mpc_baseline</code>	0.107 ± 0.083	0.861	0.059	0.283	0.583
<code>distance_threshold_graph</code>	0.107 ± 0.083	0.871	0.055	0.285	0.581
<code>last_contact_persistence</code>	0.107 ± 0.083	0.860	0.053	0.311	0.609
<code>pairwise_contact_classifier</code>	0.107 ± 0.083	0.541	0.235	0.270	0.665
<code>state_only_dynamics_model</code>	0.107 ± 0.083	0.904	0.043	0.297	0.596
<code>topology_world_model</code>	0.107 ± 0.083	0.610	0.209	0.158	0.716

Table 1: Combined-stress results. Success is the downstream task criterion. Edge F1 and graph edit are intermediate topology diagnostics.

The strongest non-oracle success baseline is `ensemble_uncertainty_planner`. The paired topology-minus-ensemble success difference is -0.018 ± 0.035 . This fails the submission gate. The result is especially instructive because the topology model improves edge F1 relative to the ensemble baseline but does not improve success.

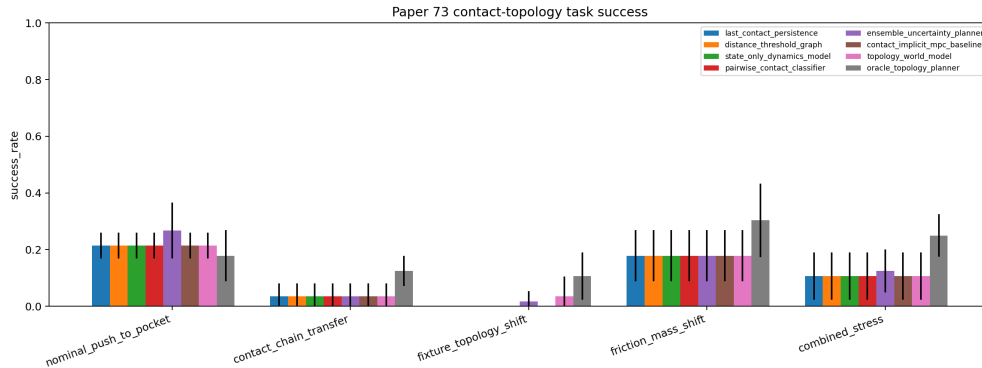


Figure 1: Task success by split. The topology model does not produce a decisive downstream success advantage.

6 ABLATIONS

Table 2 shows that topology-specific ablations are statistically overlapping. Removing the uncertainty penalty slightly improves mean success, while the full model does not separate from no-birth/death, no-component, or no-jam variants. The mechanism is therefore not supported.

Ablation	Success	Edge F1	Graph edit	Safety
topology.no-uncertainty-penalty	0.095 ± 0.097	0.590	0.221	0.116
topology.full	0.071 ± 0.097	0.583	0.220	0.081
topology.no-birth-death	0.071 ± 0.097	0.569	0.224	0.170
topology.no-component-head	0.071 ± 0.097	0.589	0.218	0.081
topology.no-jam-slip	0.071 ± 0.097	0.554	0.225	0.061
topology.no-topology-planner	0.071 ± 0.097	0.598	0.213	0.230

Table 2: Combined-stress ablations. No topology component produces a decisive task-success gain.

7 STRESS SWEEP AND NEGATIVE CASES

The stress sweep shows the same pattern: topology sometimes improves graph diagnostics but does not dominate task success. The oracle remains far from perfect at high stress, which means the local benchmark is hard but not yet a polished external benchmark contribution. Negative cases concentrate around fixture-induced graph changes and wall/fixture jams where the topology predictor identifies contacts but the planner still chooses poor contact chains.

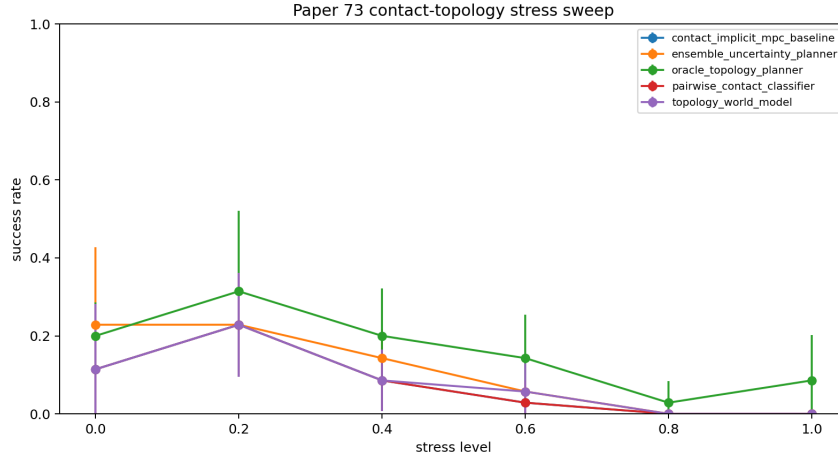


Figure 2: Stress sweep success. The topology model does not dominate the strongest non-oracle baselines.

8 RELATED WORK PRESSURE

The hostile prior-work set is crowded. Visuotactile world modeling (OmniVTA2026), robustness studies for world-action models (WorldActionModelsRobustness2026), tactile/diffusion contact-rich policies (PolyTouch2025), contact-implicit MPC (ContactImplicitMPC2025), hybrid force-position policies (ForcePolicy2026; ForceVLA2ContactRich2026), and adaptive contact-rich assembly (ContactRichAssemblyReprogramming2026) all put pressure on the novelty boundary. In this context, a contact-topology representation must deliver downstream gains. It does not in the current evidence.

9 LIMITATIONS

This rebuild is not a real robot paper. It lacks hardware experiments, videos, public benchmark validation, and a learned topology model trained on large contact-rich datasets. The local MuJoCo benchmark is simplified and the oracle success rate is low, indicating that task calibration remains imperfect. These limitations would matter even if the method won; because it does not win, they reinforce the archive decision.

10 CONCLUSION

Paper 73 should not be submitted to ICLR main. The v4 rebuild provides useful evidence, but the evidence is negative: contact-topology prediction improves some graph metrics without improving downstream contact-chain control. The correct terminal state is **KILL/ARCHIVE**.

REFERENCES

- ContactImplicitMPC2025. Contact-implicit model predictive control: Controlling diverse quadruped motions without pre-planned contact modes or trajectories, 2025. <https://doi.org/10.1177/02783649241273645>.
- ContactRichAssemblyReprogramming2026. Fast reprogramming and adaptive reproduction of contact-rich assembly, 2026. <https://doi.org/10.3389/frobt.2026.1746577>.
- ForcePolicy2026. Force policy: Learning hybrid force-position control policy under interaction frame for contact-rich manipulation, 2026. <http://arxiv.org/abs/2602.22088v2>.
- ForceVLA2ContactRich2026. Forcevla2: Unleashing hybrid force-position control with force awareness for contact-rich manipulation, 2026. <http://arxiv.org/abs/2603.15169v1>.
- OmniVTA2026. Omnivta: Visuo-tactile world modeling for contact-rich robotic manipulation, 2026. <http://arxiv.org/abs/2603.19201v2>.
- PolyTouch2025. Polytouch: A robust multi-modal tactile sensor for contact-rich manipulation using tactile-diffusion policies, 2025. <http://arxiv.org/abs/2504.19341v1>.
- WorldActionModelsRobustness2026. Do world action models generalize better than vlas? a robustness study, 2026. <http://arxiv.org/abs/2603.22078v3>.